

Institute Technical Summer Projects (ITSP)

Project Reference Documentation

Team Name	Project Apex 4
Project Title	Dental Spectrophotometer
Domain / Category	Medical devices
Mentor(s)	Animesh Jain

Team Members

List every team member with their official roll number (format: 25BXXXX).

S.No.	Name	Roll Number	Role / Responsibility
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Project Details

2.1 Problem Statement

Getting the right tooth color is an important step when making crowns, veneers or fillings. If the color is a little bit off from the other teeth it is easy to see. A lot of times the dentist has to redo the work because the color's not right. In dental offices in India the dentist still matches the tooth color by eye. They use a tool called a VITA shade guide to help them choose the right color.

This way of matching colors is not very accurate because it depends on the dentist's ability to see colors. If the dentist is tired or the light in the room is not right they might choose the color. Sometimes two dentists will not agree on the color for the same tooth. There are tools called spectrophotometers that can measure the tooth color accurately.. These tools are very expensive around ₹1–2 lakh, which is too much money for smaller dental offices.

Because of this not many dentists in India use these tools. A lot of dentists do not want to be the first to try a tool that is not well known. This affects both the dentists and the patients. The dentists have to pay for the work to be redone. They might get a bad reputation. The patients have to deal with teeth that do not look right they have to spend time in the dentists chair and they might have to pay more money.

This project is trying to fix a problem. There is no tool that's affordable and accurate for smaller dental offices in India to measure tooth color. Tooth color measurement is very important for tooth restorations like crowns,

veneers and fillings. Tooth restorations like crowns, veneers and fillings require tooth color matching. The right tooth color is necessary for tooth restorations, like crowns, veneers and fillings.

Why each existing solution falls short

Visual shade guides (VITA Classical / 3D-Master). Inexpensive and ubiquitous, though subjective. Human color perception proves to be inaccurate in discriminating colors accurately in varying lighting conditions, is prone to metamerism (i.e. two samples appear to be matching under one light source and not under another), and tends to become inaccurate due to fatigue of the observer when comparing the two samples side by side.

Commercial spectrophotometers / colorimeters. Accurate and objective but costly (₹1-2 lakh) and often bulky. It is the cost factor, not the technology itself, which proves to be the critical limiting factor in the target clinics.

Intraoral scanners (IOS) with colour mapping. Increasingly used in upmarket clinics but specifically designed to obtain 3-D geometry and not calibrated colour. Since the angle and distance of illumination keep changing during scanning, there is no control on reflection effects and colour calibration is done by a 3-channel RGB camera. Thus, they cannot replace objective colorimetry in difficult aesthetic situations

2.2 Solution Proposed

Core mechanism / working principle

The concept device is a portable, self-contained spectrophotometer based on 45°/0° illumination/detection configuration and a cheap multi-channel spectral sensor. This design concept is based entirely on the optical properties of a tooth. When white light falls onto a tooth surface, it separates into two streams. About 4-5% of it reflects off the thin layer of saliva covering the enamel in a mirror like fashion: this reflection has the same colour as the lamp light and does not provide any information about the colour of the tooth itself, thus must be filtered out. Another roughly 95% of light goes through the translucent enamel, enters the dentin layer and scatters through the complex structure of microscopic tubules. During the scattering inside the dentin, the light gets absorbed by organic pigments and hydroxyapatite while the long wavelength light is reflected back to the surface. Thus, it is the light from the diffuse reflection that needs to be measured.

The geometric design of the optical system is such that the useful diffuse light is retained while the unwanted specular glare is filtered out. Two LED sources with a high CRI and daylight equivalent color temperature (~5000 K) are used to illuminate the tooth at an angle of 45° with respect to the collection axis. As the angle of incidence equals the angle of reflection for the specular component, the glare is reflected at an angle such that it does not fall in the collection path and only the diffuse light from the tooth, scattered uniformly in all directions, reaches the collection path at 0°.

Then the collected diffused light goes through an optical system – a light-tight chain made of entrance aperture, spatial-filter pinhole, baffled (ridged) focal tube, achromatic collimating lens, infrared cut-off filter, and diffuser before getting into the AS7341 spectral sensor where the light intensity is reported for eight visible band channels (plus Clear channel, near-infrared channel and flicker detection channels). After that, the firmware and the host software perform the necessary calculations and convert the results from eight bands to a device-independent color and compare it to the nearest standard dental shade. The most significant concept involved in the entire process is relative reflectance normalization: each measurement procedure involves taking three sequential measurements – a dark measurement (LED off/probe blocked), white-reference measurement (probe on a

calibrated white tile), and the tooth measurement. These three readings are combined together in such a way that the measured color becomes independent of LED brightness drifts, sensor gain, temperature, and ambient light. As a result of that, two different devices measuring the same tooth give the same color value.

Key subsystems / modules

1. Standardised illumination + constant-current LED driver

There are two high-CRI surface-mount LEDs with a ~5000 K colour temperature mounted in the probe tip at an angle of 45° providing a stable daylight-like spectrum so as not to depend on ambient light in the clinic. The most important feature here is the use of a constant current source, i.e., not to connect LEDs directly to the power supply since the LED powered directly by a battery will change its current and therefore brightness and even spectrum depending on the supply voltage and junction temperature thus changing the very reference point of measurements. This constant current driver is made using the LT1490 op-amp, IRLZ44N N-channel MOSFET, and current sense resistor as shown in §2.3. The current is kept deliberately low in order not to raise the junction temperature and thus keep the spectrum stable.

2. Precision optical collection & stray-light-rejection barrel

This is the physical core of the measuring device: an airtight and lightproof housing, which captures, filters, and optimizes the dispersed light before reaching the light sensor. The following are its specifications:

Entrance aperture ($\varnothing 5.0$ mm) is smaller than a maxillary central incisor (width 8.0-8.5 mm), enabling the device to measure only the middle third of the tooth and not include the pink-colored gingiva or the translucent incisal edge.

Alignment dish (diameter = 13.82 mm) that touches the tooth's surface, ensuring no light from the surrounding room enters the system through this gap.

Spatial filter pinhole (500 μm or 0.5 mm diameter), which is located right after the aperture and well before the lens. This serves to cut off high-angle beams and serves as a point source for the next lens to collimate.

Ridged focal tube with a constant 60° inner V-groove (baffles) and very matte black inner coating (less than 2% reflectance). The edge diffraction from the pinhole results in grazing angles of light, which are captured by multiple inner reflections and absorbed by the coating, since even the black SLA resin is capable of reflecting about 5–10% of light under such conditions.

3. Custom achromatic optics

An achromatic optical configuration is used to collimate the divergent cone of light emanating from the pinhole into parallel rays without any wavelength-dependent shift in the focal position, which is absolutely necessary since all eight spectral bands have to be focused at the same level. To keep costs low and avoid minimum order quantities (§2.3), the system makes use of two doublets of catalogue lenses with focal lengths of approximately 25 mm (convex BaF10, concave SF10) and 11 mm (convex SK4, concave F2), respectively.

4. Spectral conditioning + sensor interface

After the collimation process, the parallel beam is subjected to an IR cut filter (which cuts out the near infrared wavelengths above 700 nm since the silicon sensor will read these as noise; it is positioned in the parallel beam such that the interference coating receives incident light from it at about 0 degrees and does not undergo blue shifting), followed by a diffuser (to diffuse the beam so that all channels on the sensor receive the same average spectral information with no channel reading a hot spot), and lastly the AS7341. Due to the fact that the active silicon is 1.85 mm below the top surface of the sensor, a stepped plunger of height 0.85 mm is extended into the sensor to ensure that the diffuser is kept at an exact 1.00 mm optical distance away from the silicon.

5. Readout electronics + embedded processing

The AS7341 sends signals through the I²C protocol to the microcontroller, which controls the sequencing of the LED light, initiates the three-part process of measuring dark, white, and tooth data, and transmits data to either the PC or the phone via USB and Bluetooth.

6. Software pipeline + GUI

The four-step colour pipeline, which is fully explained in §2.3, processes the eight channel readings and outputs a CIE Lab* colour value, which is compared to the 32 VITA colours using the CIE76 ΔE Euclidean distance to output the three best matching colours with their corresponding ΔE distances. The desktop GUI checks the correct order of dark → white → tooth colours and operates in two modes: simulation (values generated in the application, no hardware involved) and live (serial connection).

What differentiates this from existing solutions

Cost. A bill-of-materials target costing under ₹10,000 – about ten times cheaper than the ₹1-2 lakh commercial devices – that is why the target clinics can utilize it.

Commercial optics. The use of catalog optics (BaF10/SF10, SK4/F2) as opposed to achromatic optics that are made requires at least 500–1000 pieces and takes several weeks, making it impossible to afford at prototype level.

Diffuse Isolation. The 45°/0° geometry makes it impossible for the device to suffer from specular glare as opposed to an IOS where illumination angle and standoff distance keep changing during scanning process.

Spectral, not RGB. Eight spectral bands provide accurate lightness L and yellowness b measurements instead of an approximation and blurring of 3-channel RGB camera.

2.3 Work / Simulations Completed Till Now

The work completed so far includes research and simulations in the following four aspects: the literature study and design study; the optomechanical CAD design; the optical component and purchasing study; the electrical design study together with its LTspice simulation verification; the color-processing pipeline design; and the market study for clinical application. Fabrication (by 3D printing), assembly, and clinical evaluation remain to be done and the plan is included in the timeline (§2.4). Neither FEA/thermal simulation nor physical prototype has been completed before so far, therefore none of them can be claimed – the only simulation that has been done is the electrical simulation by LTspice (item 4).

1. Literature review / design study

1.1 Standards of colorimetry and color measurement process

The whole project is based upon the colorimetry system defined by the CIE, since the value of the device lies in its capability to output a standardized, internationally comparable color value, but not just a subjective judgment. The team conducted a detailed study in the following areas: the CIE 1931 2° Standard Observer Color Matching Functions (the table of response of the average human eye on each wavelength), the D65 standard illuminant and its white point ($X_n = 95.047$, $Y_n = 100.000$, $Z_n = 108.883$), CIE Lab* color space standardised in ISO/CIE 11664-4, the CIE76 ΔE color difference metric. In addition to the above, the clinically accepted thresholds of color difference have been adopted as the project's goal of accuracy: a ΔE of under 1.0 is imperceptible by human eyes; ΔE of between 1.0 and 3.0 is clinically acceptable for the color of a restoration; and a ΔE of above 3.0 shows an insufficient match and must be remeasured.

1.2 The optics of natural teeth

A tooth is not just one solid-colored thing but a multilayered translucent material; understanding the structure of that material is exactly what dictated the illumination geometry of the device. The outermost layer of functional tooth material is a layer of saliva over the crystalline enamel; the optical properties of such a layer resemble those of a layer of glass, and about ~4–5 % of light is reflected specularly from this layer. Specular reflection reflects the color of the light itself, and has nothing to do with the tooth's color, so this reflection should be ignored. Underneath the enamel there is the dentin, which is dense and porous tissue filled with tubules; it is into this tissue that the rest (~95 %) of the light enters. The light undergoes the process of multiple scattering inside the dentin, during which pigments of the dentin selectively absorb blue and violet light and reflect yellow and red light, hence giving natural teeth the warm color. The practical implication is that specular reflection follows “angle in equals angle out”, so 45° illumination makes the glare fall off the axis of the light collecting system while diffused light coming from the dentin still gets there. The ranges of natural teeth colors derived from the literature ($L \approx 70\text{--}85$, $a \approx 0\text{--}6$, $b^* \approx 10\text{--}25$) were used as sanity bounds.

1.3 Spectral-sensor technology survey and the choice of the AS7341

Choosing the detector was a critical point in the design process because not only it defined what spectral data can be acquired but also had a significant impact on the cost and firmware complexity. The team compared the AMS AS7341 against the more powerful AS7265x family of detectors and against the general type of dispersive spectrometers.

The AS7341 is an integrated eleven-channel spectral sensor: eight of its channels (F1–F8) are sensitive to light with the wavelength of 415 nm to 680 nm, and the other three channels include a clear (unfiltered) channel, a near-infrared channel, and a flicker detection channel. The device is a tiny, cheap single chip with a simple I²C

interface. The AS7265x, on the contrary, is a three-chip “triad” (AS72651/AS72652/AS72653) and has eighteen channels covering the wavelength range of 410 nm to 940 nm (that is, the visible band and a part of the near-infrared band).

This was a choice based on two arguments. The first one is economic, since the AS7341 sensor is significantly cheaper than the AS7265x sensor triad. This is the fundamental parameter that defines the entire project. The second reason is equally relevant – the relevance of the spectral ranges. The dental shade is a visible light color quantity, and the CIE Lab* calculation algorithm performed by the device uses the reflectance values only from the approximate range of 400-700 nm. The majority of the channel budget of the AS7265x and its cost are taken up by the near-infrared ranges, which do not have any information on tooth color and will just be thrown away. Eight visible channels of the AS7341 are enough for fine reconstruction of the visible reflectance curve and integration against the CIE color matching function for determination of VITA shades. Buying the AS7265x NIR range would have been meaningless and expensive both in terms of money and firmware development for no colorimetric gain.

Detector	Channels / range	Interface & size	Cost	Verdict for this project
AS7341	11 total; 8 visible (~415–680 nm) + clear/NIR/flicker	Single chip, I ² C, small	Low	Chosen — visible bands sufficient for Lab*; cheapest fit
AS7265x (triad)	18 channels, ~410–940 nm (visible + NIR)	3 chips, larger board	Higher	Rejected — NIR bands wasted for tooth colour; costlier/complex
Dispersive spectrometer	Continuous, high resolution	Bulky optics + detector array	High	Rejected — too large and too expensive for a handheld unit

1.4 Illumination survey and choosing a high-CRI LED

The lighting system was considered equally thoroughly, since spectrophotometer can only measure reflectance in wavelengths the light emitted by the illuminant. CRI is the crucial metric here – a high-CRI white LED will have a full, relatively smooth spectrum similar to the daylight, while a regular low-CRI LED will have a clear lack of blue-green frequencies (around 480 nm) and insufficient red output. If there is no emission from the light source in certain wavelength range, the reflectance measured in this band will be entirely noise-dominated and hence unreliable. Therefore, the necessity of using a high-CRI illuminant is a prerequisite for accurate colour measurement, thus a ~5000 K daylight-matched, high-CRI LED had to be chosen instead of a regular white LED.

Here is a survey of three possible high-CRI sources:

Nichia Optisolis (3030) – CRI 98. The most suitable from optical standpoint, with the flattest spectrum resembling the daylight. This LED was not chosen due to unavailability in prototypes quantities, being an industrial-grade component produced in bulk quantities.

Seoul SunLike (3030) – CRI 95-98. Also great from the optical standpoint, but this particular part was not available to the group at the time of procurement.

Lumileds LUXEON 3535 – CRI 90-95. The chosen one. While having slightly lower CRI than the other two, it is the best-performing high-CRI LED available in required small quantity.

Hence, the choice of a source is a procurement-based compromise: the Optisolis would have been the most optically suited, but the LUXEON 3535 is the highest-CRI daylight LED that was actually procured and has sufficient CRI to provide a spectrum full enough for measuring reflectance.

LED candidate	CRI	Status	Reason
Nichia Optisolis (3030)	98	Eliminated	Industrial part — bulk order only, unavailable at prototype scale
Seoul SunLike (3030)	95–98	Eliminated	Unavailable to the team at sourcing time
Lumileds LUXEON 3535	90–95	Selected	Highest-CRI daylight LED actually procurable in small quantity

1.5 Optical-glass and achromat theory

Lastly, the team analyzed the classical achromat theory to prove the choice of lenses used (the actual component choice is described in item 3). One cannot use a simple lens without chromatic aberration because the refractive index of the glass is dependent on the wavelength; hence, blue light will be deflected more and focused at a different spot, thus resulting in color fringing. The solution to the problem is an achromatic doublet, which glues or spaces apart a positive low-dispersion crown glass lens with a negative high-dispersion flint glass lens such that the chromatic aberrations of the two lenses at two different wavelengths nullify each other. Dispersion can be expressed using the Abbe number $V_d = (n_d - 1)/(n_F - n_C)$, where n_d , n_F , and n_C are the refractive indices at the d-, F-, and C-lines. A high Abbe number implies a low dispersion (like crown glasses) and vice versa. This theory is the reason why the specific glass pair used in item 3 (a high refractive index barium/dense flint pair followed by a crown/flint pair) is correct to focus the eight AS7341 bands on one focal plane.

2. CAD models — completed in Autodesk Fusion 360

The entire opto-mechanical system was designed parametrically in Autodesk Fusion 360, starting from the end of the probe towards the sensor along the optical axis. The mechanical elements were selected keeping in mind the optics and not just to join things together, and all of them were parametric so that clearance tuning could be done post initial fitting.

2.1 Probe tip (Ø13.84 mm dish, Ø5.0 mm aperture, dual 45° LED ports)

The probe tip was represented by a conical dish in order to ensure its automatic alignment and sealing with respect to the tooth structure, thus preventing environmental light from entering the contact point. The Ø5.0 mm central opening limits the area under analysis to the middle one-third of a central incisor, thereby excluding the light contribution from the pink gingival margin and the clear incisal edge. The two illuminating ports are tilted at 45° relative to the detection axis.

2.2 Spatial-filter pinhole plate (Ø0.5 mm)

A 500 µm pinhole plate is located between the probe tip and the focusing tube, but before the lens itself. This acts as the point source for the system, removing stray rays which are either off-axis or of high angles, and creating the nice, small cone of light to be used by the optics.

2.3 Ridged focal tube (Ø11.4 mm bore, 60° V-grooves, 0.40 mm clearance)

The focal tube retains the Ø11 mm mating tube in a receiving bore designed at Ø11.4 mm, thus creating an intentional clearance of 0.40 mm (0.20 mm on each side). Such clearance was also a design decision. An 0.80 mm clearance was ruled out due to excessive looseness which would allow for optical tilt and light ingress; a clearance

less than 0.30 mm would have resulted in the binding of the tube on the layering lines caused by FDM printing. The 0.50 mm $\times$ 45° chamfer facilitates smooth insertion of the tube without any plastic removal, while the 0.25 mm clearance at the bottom makes sure that the tube rests against its shoulder rather than bottom out on any print artifacts. The tube is locked in place by radial set screws or pins drilled into a wall of 12-13 mm diameter (1.5-2.0 mm thick), being sufficient for not allowing the thread to strip. The internal bore features continuous 60° V-grooves which serve as light traps to be later coated with matte black as described in §2.2 of the solution.

2.4 Sensor plunger adapter (0.85 mm step → 1.00 mm optical gap)

The most critical element with respect to tolerances in the assembly process is the sensor interface. Given that the active silicon is 1.85 mm away from the top surface of the package housing the AS7341 sensor, placing a diffuser on the top surface of the sensor package results in a clearance of 1.85 mm between the diffuser and the active silicon. A stepped plunger, which penetrates 0.85 mm into the cavity created by the sensor package and a pocket at the end of which carries the diffuser, ensures that the diffuser is positioned 1.00 mm away from the active silicon.

2.5 End caps and recorded design iterations

The snap-fit end caps prevent the intrusion of ambient light while making sure that the module can be serviced. Three design decisions were made while working with the model. First, the size of the entrance aperture was decreased incrementally from 10 mm to 6 mm to about 3.5-5 mm while trying to get an accurate fit inside the oral cavity in relation to the width of the incisors. Secondly, the focal-tube gap was set according to the above compromise of 0.40 mm. Thirdly, the optical components sequence was decided upon after the first disagreement of two designs, where the final choice became collimate → IR-cut → diffuse → sense.

3. Optical component selection & procurement study

A hard supply chain constraint influenced the optics selection. Precision custom achromats are typically supplied with a minimum order quantity of 500 to 1000 units and long delivery times; purchasing custom-ground lenses in single-digit quantities would make each lens very expensive, besides introducing additional complications in form of import duties. Since all components of the telescope need to cost less than ₹10,000, the custom optics solution was eliminated, and instead, the telescope had to be built using standard lenses that can be purchased one at a time. The four lenses used are the least expensive combination possible for the team in single units that still performs an achromatic function, and their glass is also better optically than other stock alternatives considered.

The optics are arranged as a doublet of achromatic lenses, with visible BBAR coating to account for the wavelength shift in blue region in the case of no coating.

Element	Glass	Type	nd (index)	Vd (Abbe)
Convex	SK4	Dense crown	1.6127	58.63
Concave	F2	Flint	1.6200	36.37

The crown and flint elements—SK4 ($n_d = 1.6127$) and SF2 ($n_d = 1.6477$)—possess higher refractive indices than ordinary crown glass ($n_d \setminus 1.517$). This higher index bends light more sharply, allowing the target focal length to be achieved over a shorter axial distance with reduced spherical aberration—a distinct advantage in a compact handheld barrel where length is constrained.

4. Electrical design + LTspice simulation

Unlike the other subsystems that have not yet been simulated in LTspice, simulation of the LED driver subsystem has been done prior to building the hardware since it controls the stability of the light. Since all readings depend on the white tile readings, any variation in the light between the white and tooth reading leads to color error.

4.1 Why a constant-current driver is necessary

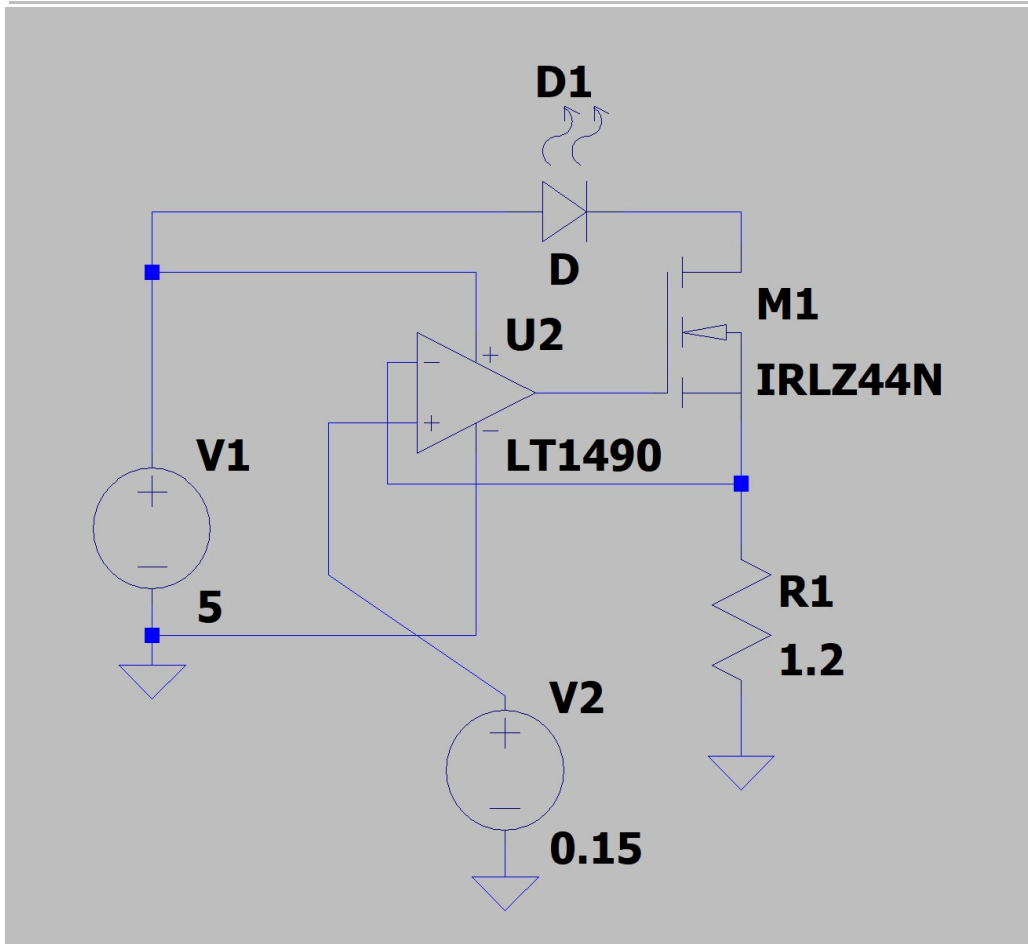
It is not possible to drive an LED using fixed voltage source because the forward current varies exponentially with forward voltage. And as the temperature of the junction varies, so does the variation in the relation. This means that using fixed voltage source, the current and hence the intensity of light emission will vary with temperature and source variation. A simple linear voltage regulator will not work for the same reason because even a small variation in voltage of about 0.1V across the LED can cause large change in the current.

4.2 Circuit topology and working principle

The control circuit comprises a typical op-amp/MOSFET current sink circuit, which consists of three active components: LT1490 op-amp, IRLZ44N logic level N-channel MOSFET, and a $1.2\ \Omega$ sense resistor (R1). The LED (D1) and the MOSFET are wired in series with R1 so that all the LED current passes through the sense resistor. A reference voltage of 0.15 V is applied to the non-inverting input of the op-amp, while the voltage across R1 is fed to the inverting input of the op-amp. The op-amp controls the voltage at the gate of the MOSFET and, by means of negative feedback, changes the voltage at the gate until the voltages on both inputs become equal, meaning that the voltage across R1 will be equal to the reference voltage of 0.15 V. Since the voltage across the sense-resistor is fixed, the LED current is fixed to:

$$I(LED) = V(ref) / R(sense) = 0.15\ V \div 1.2\ \Omega = 0.125\ A = 125\ mA$$

This particular reference is established using the potentiometer in the circuit prototype that generates the 0.15 V, so that the operational current is trimmed while searching for the required illumination level empirically; after finding the optimal current in relation to measurements taken, the potentiometer will be substituted for a fixed voltage divider. IRLZ44N was selected due to being a logic-level MOSFET whose gate can be fully turned on through the rather small voltage provided by the op-amp, and LT1490 due to being able to generate the rail-to-rail output that is close enough to the rails in order to turn the gate on; this rail-to-rail property is not possessed as well by a standard LM358 (mentioned in the earlier version).



4.3 Simulation setup and results

Transient analysis (.tran 10 ms) was conducted from the main supply source of 5 V DC (V1). Two quantities were plotted: the LED current I(D1) and the supply node voltage V(n001). The outcome verifies the circuit's design:

LED current I(D1) stabilizes almost immediately at about 124.95 mA – close to the 125 mA specification within 0.1% accuracy (the measurement displays 124.9455 mA at 4.56 ms), illustrating that the feedback circuit controls the current precisely where the sensing resistor and the reference demand it.

Supply voltage V(n001) remains perfectly flat at 5.000 V over the whole 10 ms span with no sag evident under the load, indicating that the circuit offers a good load and that the regulation is not struggling against supply sag..

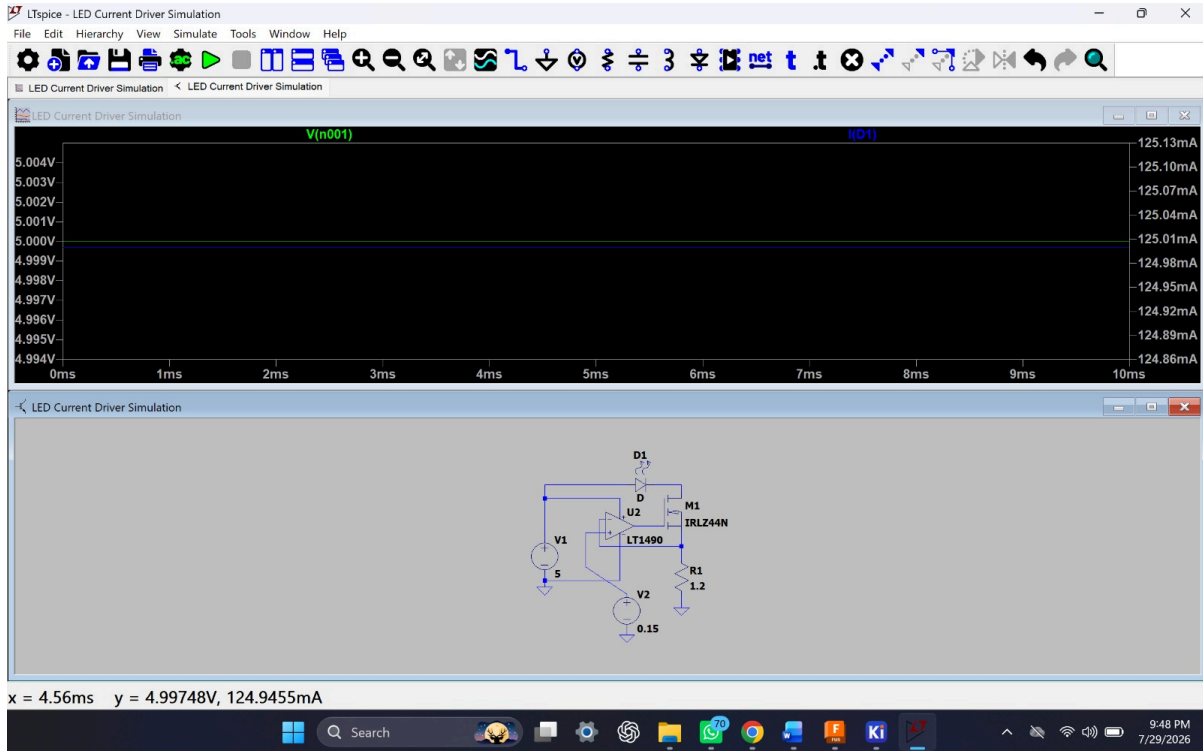


Figure 1. LTspice constant-current LED driver — schematic (LT1490 op-amp, IRLZ44N MOSFET, $R1 = 1.2\ \Omega$ sense resistor, $V_{ref} = 0.15\text{ V}$) and 10 ms transient showing $I(D1)$ locked at $\sim 124.95\text{ mA}$ with $V(n001)$ steady at 5.000 V .

This ensures validation of the governing equation and stability of the control loop without soldering any component on the board, thus making the driver a completed sub-system design. 125 mA value is considered a temporary one until the optimal illumination current level is determined during bring-up of the hardware and finally set using the final resistor divider.

4.4 Power and thermal considerations

Running at a modest current rather than the LED's maximum keeps junction temperature down and the spectrum stable. With a $\sim 5\text{ V}$ rail, a typical LED forward drop of $\sim 3\text{ V}$, and 0.15 V across the sense resistor, the MOSFET drops the remaining $\sim 1.85\text{ V}$ and dissipates on the order of $1.85\text{ V} \times 0.125\text{ A} \approx 0.23\text{ W}$, while the sense resistor dissipates only $I^2R \approx 0.125^2 \times 1.2 \approx 0.02\text{ W}$ — both comfortably small. The firmware further limits heating by energising the LEDs only during an actual reading rather than continuously.

5. Software / algorithmic pipeline — implemented and validated in simulation

The full pipeline for the color processing has been achieved (using Python code) and tested using the simulation mode within the GUI so that the algorithm has been validated independent of the hardware. The eight values from the sensors are transformed to the standard dental shade in four steps.

5.1 Stage 1 — relative reflectance normalisation

Counts generated by the AS7341 raw integers mean nothing, since their values depend on LED illumination intensity, the gain, integration time, temperature, and the presence of ambient light. There are three measurements made per channel: the dark measurement (noise floor of the sensor and stray light), white reference measurement (how the known white tile responds under the current hardware settings), and tooth measurement. :

$$R(\lambda) = [R_{\text{tooth}}(\lambda) - R_{\text{dark}}(\lambda)] / [R_{\text{white}}(\lambda) - R_{\text{dark}}(\lambda)]$$

This results in eight reflectance values ranging from 0.0 to 1.0, whereby 0.0 reflects no reflection while 1.0 equals the reflectance of the reference white tile. This is because the dark value is subtracted from both numerator and denominator and the ratio obtained has the same units when measuring the same tooth; guards restrict the value to within the correct range and avoid division by zero.

5.2 Stage 2 — CIE XYZ tristimulus conversion

The eight reflectance values are then mapped into the way the human eye actually perceives colour, since the eye is far more sensitive to green (~555 nm) than to violet (~415 nm) and a naive sum of channels would distort the colour. Each reflectance value is weighted by the CIE 1931 2° Standard Observer colour-matching functions and the D65 illuminant spectral power distribution $S(\lambda)$, and summed across the channels:

$$X = k \cdot \sum R(\lambda) \cdot \bar{x}(\lambda) \cdot S(\lambda) \quad Y = k \cdot \sum R(\lambda) \cdot \bar{y}(\lambda) \cdot S(\lambda) \quad Z = k \cdot \sum R(\lambda) \cdot \bar{z}(\lambda) \cdot S(\lambda)$$

Here Y is the perceived luminance and the scaling constant k normalises the white reference so that Y = 100. The output is the three tristimulus values (X, Y, Z) describing the colour in a device-independent, perception-weighted space.

5.3 Stage 3 — CIE Lab* conversion

Tristimulus values will be transformed to the CIE Lab* color space, which is the standard system for dental color communication all around the world, where equal differences in numbers imply equal visual differences in colors. X, Y, Z tristimulus values will be normalized to the white point D65 (X_n , Y_n , Z_n), and then the following transformation function f(t):

$$L = 116 \cdot f(Y/Y_n) - 16 \quad a = 500 \cdot [f(X/X_n) - f(Y/Y_n)] \quad b^* = 200 \cdot [f(Y/Y_n) - f(Z/Z_n)]$$

L is lightness from 0 (black) to 100 (white); a is the red–green axis (positive = red); b is the yellow–blue axis (positive = yellow). Natural teeth fall in the ranges $L \approx 70$ –85, $a \approx 0$ –6 and $b \approx 10$ –25, the warm, light region expected of dentin. The output of this stage is the compact three-number vector [L, a, b*] that fully identifies the measured tooth's colour.

5.4 Stage 4 — ΔE shade matching

Finally the measured [L, a, b] is compared against a stored database of the 32 VITA Classical and 3D-Master shades, each held as its published Lab coordinates, using the CIE76 colour-difference formula — the straight-line (Euclidean) distance between two colours in Lab* space:

$$\Delta E = \sqrt{(L_d - L_r)^2 + (a_d - a_r)^2 + (b_d - b_r)^2}$$

All 32 shades are ranked by ΔE and the three closest are returned with their scores and a quality label, applying the thresholds from item 1.1 (ΔE < 1.0 excellent, 1.0–3.0 acceptable, > 3.0 re-measure). This is what turns a physical measurement into an actionable dental shade code with no human judgement involved.

5.5 Graphical interface and dual-mode operation

A GUI for a desktop computer incorporates all four stages in a single clinical process. A sequential state machine ensures the dark → white → tooth process and only enables the next action once the prior step is done, ensuring that the operator does not perform an un-calibrated measurement. On receiving the reading for a tooth, the GUI implements the whole pipeline and shows the eight-channel reflectance fingerprint, the selected shade with the associated ΔE and quality grade, as well as the three most similar shades with a rough color sample. A single toggle switch selects the simulation mode, which sends pre-approved dummy input when buttons are pressed, allowing the pipeline to run without any hardware connected, and the real mode, where a separate thread monitors the USB serial connection and feeds any input coming from the microcontroller through the pipeline in exactly the same way.

6. Market / clinical survey (summary)

A quick market survey was conducted for 12 dental practices to verify the demand and determine the realistic target customers. The dental practices could be grouped into three classes, namely, Grade A high-end dental clinics (about 150 patients per month), equipped with in-house CAD/CAM systems and monolithic zirconia workflow, whose interest is low because their current digital technologies meet their needs; Grade B mid-tier dental clinics (about 50 patients per month), utilizing intraoral scanners along with visual shade guides, whose interest is moderate and conditional upon rapid calibration process and simplicity of interface; Grade C low-end dental clinics (about 20 patients per month), using only manual VITA shade guides, whose interest is high since they realize the limitations of visual matching yet cannot afford ₹1-2 lakh imported equipment. Thus, it is evident that Grade C (and to a certain extent Grade B) dental clinics comprise the target market, and their key reluctance is the lack of peer review publication of the technology, meaning that one should carry out ΔE validation study together with a partner clinic/college as the next move to market.

Key results / design decisions arrived at

Optical principle fixed: a $45^\circ/0^\circ$ illumination-and-collection geometry, chosen from the tooth-optics study, rejects the 4–5 % specular glare and isolates the ~95 % diffuse dentin light that carries the true shade.

Detector selected: the AS7341 (8 visible bands) over the AS7265x, on grounds of cost and because the AS7265x's NIR bands carry no tooth-colour information.

Illuminant selected: the Lumileds LUXEON 3535 (CRI 90–95), the highest-CRI daylight LED procurable in small quantity after the Nichia Optisolis (bulk-only) and Seoul SunLike (unavailable) were ruled out.

Optics finalised: achromatic doublet SK4/F2 at FL ≈ 11 mm — selected to beat the 500–1000-unit MOQ on custom lenses while holding all eight bands on one focal plane.

Key CAD dimensions locked: diameter 5.0 mm sampling aperture; 0.5 mm spatial pinhole; 9.0 mm focal-tube bore at 0.40 mm clearance with chamfer and bottom relief; and a 0.85 mm plunger yielding the exact 1.00 mm sensor optical gap.

LED driver validated: the LT1490 / IRLZ44N / 1.2Ω constant-current sink was simulated in LTspice and confirmed to lock the LED current at ~125 mA ($I = V_{ref}/R_{sense} = 0.15/1.2$) with a rock-steady 5 V rail — the only formal simulation performed so far.

Software pipeline proven: the full reflectance-normalisation $\rightarrow$ CIE XYZ $\rightarrow$ CIE Lab* $\rightarrow$ ΔE matching chain is implemented and validated in the GUI's simulation mode, with $\Delta E < 1.0$ against VITA references adopted as the accuracy target for the physical build.

Market confirmed: Grade C (and Grade B) practices are the target, the sub-₹10,000 price is the decisive attribute, and a published ΔE validation study is the key to adoption.

2.4 Timeline for remaining work

<i>Phase</i>	<i>Target Timeline</i>	<i>Key Deliverable</i>
<i>Phase 2: Printing & Assembly</i>	<i>August 7</i>	<i>Iteration 2 Printed Housing & Driver Circuit</i>
<i>Phase 3: Sensor & Optics</i>	<i>August 10</i>	<i>Integrated Sensor Head (1.00 mm Gap)</i>
<i>Phase 4: Light-Tight Test</i>	<i>August 12</i>	<i>Zero-Leakage Sealed Optical Path</i>
<i>Phase 5: VITA Validation</i>	<i>August 15</i>	<i>Clinical Accuracy Report ($\Delta E < 1.0$)</i>

2.5 Expected Outcome

The end result will be a fully functional, extremely inexpensive handheld dental spectrophotometer based on an 11-channel AS7341 spectral sensor and two 5000K LED lighting sources. The device's custom opto-mechanical design uses a light tight grooved barrel, a 500 micron spatial pinhole, and an achromatic doublet biconvex lens (SK4/F2) with visible BBAR coating, which keeps the exact 1 mm distance to the sensor. The inbuilt MCU firmware takes care of processing the spectral readings by converting them to CIE L*a*b* space and computing Euclidean distances with regard to 32 VITA shade tab locations. The product operates with a 3sec chairside workflow, ensuring delta E with a manufacturing cost of less than 10000 thousand rupees. The complete release includes verified CAD drawings, open-source C/C++ code, and a benchmarking report of the target dental practices.